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UAV-assisted fair communications for multi-pair users: A multi-agent deep reinforcement learning method

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ABSTRACT

Unmanned aerial vehicle (UAV) plays an important role in scenarios like search and rescue, remote communication relay, battlefield mobile networks, etc. In this paper, we investigate multiple UAV relays providing real-time transmission for multi-pair ground users (GUs) in the absence of ground-based stations (GBSs). Due to the limited load capacities and energy resources on-board, fairness between multi-pair users, throughput maximization, as well as the connectivity maintenance between UAVs are jointly considered. We formulate the *energy-efficient fair throughput* objective function, which turns to be non-convex with hybrid variables. To solve this intractable problem, we utilize the power of neural networks in function approximation and propose a multi-agent deep reinforcement learning (MADRL)-based algorithm. Different from traditional MADRL algorithms, we utilize the method, named independent proximal policy optimization (IPPO), which allows agents to update according to their own observations memory and encourages more explorations to some extend, as the base of our solution. Simulation results demonstrate that our algorithm outperforms some baselines in terms of fairness, throughput as well as energy consumption.

1. Introduction

UAV-aided communications have been widely applied in cellular network, Internet of Things (IoT), wireless sensor network (WSN) and space-air-ground integrated networks (SAGIN). The deployment of UAVs can quickly provide communications services to GUs in situations where there is no fixed communication infrastructure on the ground (or where it is damaged, such as post-disaster areas or battlefields). Specifically, UAV-assisted applications, such as communication coverage [1], mobile relay [2], edge computing [3] and data collection [4], have become hot research topics in wireless communications and networks.

The advantages provided by UAVs are mainly summarized as follows: (1) with the progress of industrial technology, the volume of electronic devices are getting smaller while the capacities are larger, so that the capabilities of modules that can be integrated on a single UAV are enhanced with lower cost; (2) the highly controllable maneuverability of UAVs enables hazard deployments over mountains, rivers and other places that ground-based infrastructures are difficult to place, which can also realize all-terrain and seamless communications; and (3) UAVs have excellent visibility for ground communication, that is, it can reduce signal path-loss caused by obstacles and greatly improve line-of-sight (LoS) communication.

1.1. Related works

As the most widely used application, UAV-mounted base station can enlarge communication coverage and promote Quality of Service (QoS) as well as Quality of Experience (QoE) of terminals. The work in [5] studied the problem of full coverage, considering the capacity of UAVs and the QoS in terms of Signal-to-Noise (SNR) of GUs. The work in [6] studied a swarm intelligence method and their objectives were to minimize the number of deployed UAVs together with maximizing the number of covered GUs. The work in [7] assumed that an UAV was deployed to balance the heavy load of GBS and better improve the transmission rate of GUs. Based on a convex optimization method, they optimized the position of an UAV as well as bandwidth allocation between UAV and GBS to minimize the transmission delay of GUs. The work in [8] applied UAVs into the Internet of Vehicles (IoV) applications. By optimizing the altitude and data collection time of UAVs, they aimed to maximize the collected data as well as the ratio of average data to overhead. The work in [9,10] studied data collection in WSN cooperating with the trajectory design of multiple UAVs, which aims to satisfy the freshness of data in terms of Age of Information (AoI). The work in [11] studied an UAV-aided mobile edge caching

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scenario. By jointly optimizing trajectory and cache policy, they aimed to maximize the cache utility.

UAV mobile relay is another basic application in UAV-assisted communications. By establishing communication links between two or more users (or stations), UAV relays data in either instantaneous or ferrying way. The work in [12] studied the scenario, where one UAV relay serves multi-cell edge users, aiming to maximize the sum throughput of all edge users by optimizing trajectory and schedule policy. Considering an UAV-assisted relay for multi remote users, the work in [13] investigated the spectrum efficiency and power consumption issue. With the technology of non-orthogonal multiple access (NOMA), they managed to maximize the spectrum efficiency and minimize the power consumption of UAV by optimizing the trajectory of UAV. The work in [14] studied relay selection problem in IoV when multiple UAVs are available. Based on communication interrupt and transmission cost, they formulated the state transition probabilities in different time slot and managed to maximize network capacity through Q-learning. The work in [15] and [16] further studied energy efficiency and secure transmission problem in the multi-UAV relays problem in D2D network, respectively.

However, the relay-related works above only consider the transmission between user and UAV, instead of the instant transmission between users relayed by UAV, which may be practical in search and rescue as well as battle field missions, where the needs of communications are usually urgent and require reliable links. In this paper, we focus on the complete link and instant transmission between user-pair (from one user to another) through one UAV relay or multiple UAV relays if necessary. Besides, since there usually are multiple rescue crews or operation squads cooperatively performing tasks, establishing relay links for multiple pairs is of great significance to enable the mission completed timely and effectively, which has also been considered. There are few works, like [17–20], studying the user-pair serving problem. In [17], the authors maximized network throughput by optimizing UAV trajectory and users' power control. In [18], the authors promoted spectrum efficiency in D2D networks and maximized network throughput with the same optimized variables as in [17]. Apart from throughput, the work in [19] also investigated the energy consumption problem. In [20], the authors extended the work in [18] into multiple UAVs scenario.

As for serving multi-pair users, most of the existing works focus on the sum throughput of all pairs. This inevitably resulted in unfairness between users, where some users with good channel conditions were covered for a long time period while some remote users might be ignored to maximize the sum throughput. Relay-related works in [21–23] formulated an objective with the max–min form to ensure the minimum throughput of all pairs were maximized. The work in [21] studied the deployment of one UAV relay for multi-pair users, where the minimum rate among all pairs was maximized with a block coordinate descent (BCD) method that iteratively optimizes different variables. The work in [22] studied the deployment of one UAV relay in a two-way transmission scenario using branch-and-bound and successive convex optimization. The work in [23] studied the deployment of one UAV relay in full-duplex scenario with BCD method that iteratively optimizes user schedule and UAV trajectory. Besides, coverage-related works in [24–28] also integrated Jain's fairness index [29] into reward functions to sense and measure the fairness of services with deep reinforcement learning (DRL) based solutions, which can also achieve fair communications among multiple users. However, this has not been utilized in serving multi-pair user scenarios.

Traditionally, communication coverage needs only static deployment or slightly adjustment of UAVs to serve certain sub-area or subgroup of GUs. While in scenarios where the distribution of GUs is sparse and the whole area is relatively large, static deployment may not satisfy the requirements and the mobility of UAVs is helpful to cover more area or GUs. Hence, trajectory design is of great importance in UAV assisted researches. However, centralized approaches, like convex optimization, genetic algorithm and other heuristic methods, need global states to generate optimal solutions, which in multiple UAVs application is hard to realize due to the time-varying variables and topologies.

RL-based methods are used to solve Markov decision process (MDP) problem with continuous decision-making requirements, such as UAV trajectory design. Meanwhile, RL-based UAV assisted communications have attracted great attention of industry and academia and already been applied in dynamic access control, data rate control, wireless cache, data offloading, transmission security, etc [30]. The work in [31] and [32] studied the spectrum access and trajectory design problem with modified multi-armed bandit (MAB) algorithms, respectively. The work in [33] studied UAV-assisted data collection problem with ambiguous and deterministic area borders by DRL and multi-agent DRL (MADRL) method. The work in [34] aimed at maximizing network throughput under interference by utilizing CAA-MADDPG, the proposed algorithms based on MADDPG, to optimize the UAV trajectory and transmission power. The work in [35] studied UAV-GBS connectivity maintenance in UAV trajectory design problem with DRL. The work in [36] studied the decouple of uplink and downlink transmission. They assumed that GUs can transmit or receive from different UAVs or GBS, which aimed to maximized their throughput. With a MADRL-based method, they jointly optimized the access control of GUs and the trajectories of UAVs to maximize the sum throughput of uplink and downlink rates. The work in [37] studied task scheduling problem in SAGIN scenario, where UAV is used to collect computing tasks and make decisions on offloading with a DNN-based DRL method. The work in [38] studied packet routing problem in multi-hop relay system. Based on QMIX, they proposed MAQMIX to solve the joint connectivity maintenance, resource allocation and route selection problem.

Besides, there some DRL-enabled works pioneered in considering the fairness of services. The work in [24,25] studied the point of interest (PoI) coverage problem. Through DRL, they managed to fairly cover all PoIs during certain period with optimized trajectories. The work in [26] studied the air-ground cooperation problem with MADRL, where UAVs cooperated with GUs to maximize throughput and fairness. They further solved the hybrid action space trouble in DDPG with the transformation of discrete actions probabilistic distribution. The work in [27] utilized recurrent graph neural network to solve the integrated measurement *Connectivity-Fairness-Energy* in coverage scenario. The work in [39] also combined mean field method into DRL in fair communications. However, even though fair communications with DRL algorithms have been studied in different wireless scenarios, researches on fair communications among multi-pair users served by multiple UAVs with DRL-based methods have not been reported yet.

1.2. Motivations and contributions

Though there are a lot of works either focusing on multi-pair users, multiple UAV relays or DRL-based solutions, there are still some problems that have not been fully solved or considered. All these existing problems can be summarized as follows:

1. There are few works studying the problem of serving multi-pair users, where in emergency communications it is a common application for transmission between two specific users. Moreover, since there are multi-pair users, if we only focus on the sum throughput optimization, some of the pairs in remote distance may be ignored because their little contributions to the objective. Hence, the fairness of service among users should not be neglected.
2. The connectivity of UAVs can ensure effective cooperations. However, it is usually neglected or assumed to be maintained through back-haul links, such as satellite, which is not practical in post-disaster areas, where fixed infrastructures malfunctioned and the signals are blocked severely.

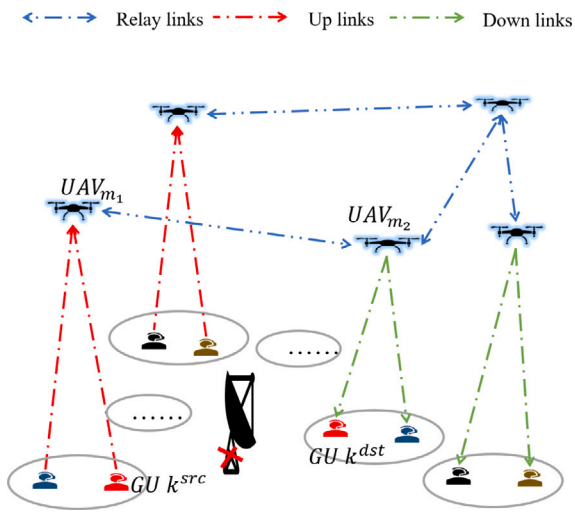


Fig. 1. System model. The different colors denote different communication pairs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3. Most of the existing works uses centralized approaches which need global state that are hard to realize without the presence of control center in multiple UAVs system. More importantly, the complexities will increase rapidly with the number of variables and iterations.

To the best of our knowledge, we are the first to simultaneously study the fairness among multi-pair users, the connectivity between UAVs and the energy efficiency of UAVs based on a MADRL approach. In this paper, we investigate the trajectories design of multiple UAVs to provide instant data transmission for multi-pair users in the absence of GBSs caused by natural disasters, as shown in Fig. 1. Considering the limited resources of UAVs and the limited communication capabilities of GUs, we aim to promote the fairness among multi-pair users and maximize energy efficiency of UAVs. Besides, in order to ensure the cooperation between UAVs and the reliability of relay links, we constrain the connectivity of UAV networks to be maintained. We propose a MADRL-based approach to solve the formulated problem. The main contributions of this paper are summarized as follows:

1. We construct the system model that multiple UAVs serve multi-pair users for fair communications. As a constraint, the connectivity maintenance of UAV networks is achieved. Moreover, we define the fairness throughput and formulate the objective function that maximize the fair throughput as well as energy efficiency;
2. We model the formulated problem as a partially observable MDP and utilize a MADRL-based method to solve the problem. Different from traditional MADRL algorithms with CTDE structure, we establish a fully distributed MADRL model, where each agent acts and updates according to their own observations instead of global states and actions. More importantly, the reward function encouraging fairness and energy efficiency is properly defined with consistency to the formulated objective;
3. We conduct extensive experiments to validate the effectiveness of the proposed approach, and analyze the performance of fairness, sum throughput as well as energy efficiency compared with some baselines.

The rest of this paper are organized as follows. Section 2 elaborates the system model. Section 3 gives problem formulation. Solutions are introduced in Section 4. Simulation results are analyzed in Section 5. Section 6 concludes this paper.

2. System model

The scenario considered is shown in Fig. 1, where multiple UAVs are connected to serve multi-pair users for fair communications in a search and rescue mission. There is a set of $\mathcal{K} = \{1, \dots, K\}$ of K user pairs with known locations acquired by GPS or remote sensing techniques. The kth pair is composed of a source GU and its destination, denoted by k^{src} and k^{dst}. The locations of the kth pair are $\mathbf{w}_k^{\text{src}} = [x_k^{\text{src}}, y_k^{\text{src}}]^T \in \mathbb{R}^2$ and $\mathbf{w}_k^{\text{dst}} = [x_k^{\text{dst}}, y_k^{\text{dst}}]^T \in \mathbb{R}^2$, respectively. During the mission, we assume source GUs keep transmitting to their destination GUs as long as relay links are established.

The mission duration is divided into a set $\mathcal{T} = \{1, \dots, T\}$ of T slots with equal length δ. Note that δ should be carefully set that the location of each UAV is assumed to be unchanged in each time slot. There is a set $\mathcal{M} = \{1, \dots, M\}$ of M UAVs deployed in the area. At time slot t, the position of UAV m is denoted by $\mathbf{w}_m^{\text{u}}[t] = [x_m^{\text{u}}[t], y_m^{\text{u}}[t], H]^T \in R^3$, where H denotes its fixed altitude that should satisfy the government's regulations or safety considerations. The maximal speed of UAV is $V_{\max}$. Hence, the movement of UAVs between two consecutive slots should satisfy

$$0 \le \left\| w_m^{\text{u}}[t] - w_m^{\text{u}}[t-1] \right\|^2 \le (V_{\max} \cdot \delta)^2, \forall m \in \mathcal{M}, t \in \mathcal{T}. \tag{1}$$

We assume all transmissions between GUs are relayed by UAVs, and no direct transmission between GUs is available due to the severely blocked signal and destroyed infrastructures. Besides, UAV adopts time division multiple address (TDMA) to serve multiple users including both source and destination GUs, and each GU can only access one UAV at each time slot. Indicator variables $\alpha_{m,k}^{\text{src}}$ and $\beta_{m,k}^{\text{dst}}$ denote whether the kth source GU or the kth destination GU is served by UAV m. If yes, their values are equal to 1 and vice versa. Hence, the access control policies should satisfy

$$\alpha_{m,k}^{\text{src}}[t], \beta_{m,k}^{dst}[t] \in \{0, 1\}, \forall m \in \mathcal{M}, k \in \mathcal{K}, t \in \mathcal{T}, \tag{2}$$

$$\sum_{m \in \mathcal{M}} \alpha_{m,k}^{\text{src}}[t] \le 1, \forall k \in \mathcal{K}, t \in \mathcal{T}, \tag{3}$$

$$\sum_{m \in \mathcal{M}} \beta_{m,k}^{dst}[t] \le 1, \forall k \in \mathcal{K}, t \in \mathcal{T}, \tag{4}$$

Apart from the blockage and remote distance of communication links, signal interference also contributes to the degradation of wireless communications. If all communication nodes operate at the same frequency, there would be severe interference to users and even cause link interruption. In order to mitigate the interference, we assume that UAVs are equipped with two transmitters operating at different frequencies when communicating with GUs and other UAVs. Moreover, since the interference management is out of the scope in this paper, we further assume each UAV communicates to other UAVs with OFDMA to eliminate the interference at UAVs.

2.1. Energy consumption model

In order to maximize the energy efficiency of UAVs, we need to quantify the energy consumption during flight. As the scope of this paper is to navigate UAVs to relay data between multi-pair users, UAVs only perform packet forwarding, hence, the communication energy consumption can be neglected compared with propulsion. Note that in some computing-intensive 5G/6G applications, where UAVs function as edge servers, the communication and computing energy consumption are comparable with propulsion. Without loss of generality, in this paper, we adopt a linear model that is proportional to the flight distance of UAVs following the work in [25] as most of the existing works have done. At time slot t, the energy consumption of UAV m is given by

$$e_m[t] = \tau_m^{\text{hover}}[t] \cdot e^{\text{h}} + l_m^{\text{move}}[t] \cdot e^{\text{f}}, \tag{5}$$

where e^{h} and e^{f} denote the units of hovering energy per second and flying energy per meter, respectively. $\tau_m^{\text{hover}}[t]$ and $l_m^{\text{move}}[t]$ represent the hovering duration and flying distance in time slot t.

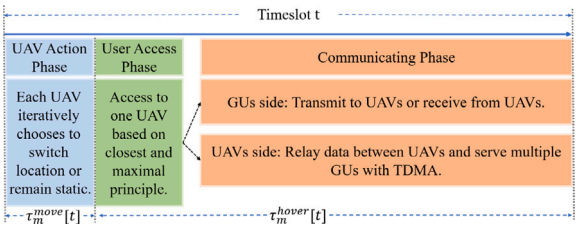


Fig. 2. Movement and Communication Protocol.

2.2. Movement and communication schedule

The movement and communication schedule is given in Fig. 2. At the start of time slot t, each UAV changes its location or remains static according to its control policy in the action phase. Then, each GU accesses to UAVs based on the closest distance and maximal SNR principle. After the access phase, UAVs receive data from source GUs or forward data to destination GUs while keeping relay data across UAV network by the separate transmitters they are equipped with. Note that UAVs will hover at the new (or old) location for the rest of the slot after action phase and communication will not occur when UAVs are moving. The duration of moving is denoted by $\tau_m^{\text{move}}[t] = \delta - \tau_m^{\text{hover}}[t]$.

2.3. Channel model

Take the kth pair served by a two-hop UAV relay link as an example, i.e., GU $k^{\text{src}} \rightarrow$ UAV $m_1 \rightarrow$ UAV $m_2 \rightarrow$ GU k^{dst}, the channel model between GU k^{src} and GU k^{dst} is defined as follows.

2.3.1. UAV-UAV channel

Channel model between UAVs follows the free space propagation model. At time slot t, the path loss between UAV m_1 and UAV m_2 is given by

$$g_{m_1,m_2}[t] = \sqrt{\frac{\rho_0}{\left\|\mathbf{w}^{\text{u}}_{m_1}[t] - \mathbf{w}^{\text{u}}_{m_2}[t]\right\|^2}}, \tag{6}$$

where ρ_0 is channel gain at reference distance (1 m), $\left\|\mathbf{w}^{\text{u}}_{m_1}[t] - \mathbf{w}^{\text{u}}_{m_2}[t]\right\|^2$ denotes the euclidean distance between two UAVs.

We assume UAV adopts orthogonal frequency-division multiple access (OFDMA) when communicating with other UAVs, hence the interference between UAVs is ignored. The SNR generated at UAV m_1 by m_2 is calculated by

$$\gamma^{\text{U2U}}_{m_1,m_2}[t] = \frac{p_{\max} \cdot g_{m_1,m_2}[t]}{n_0 \cdot b_u}, \tag{7}$$

where $p_{\max}$ denotes the maximal transmit power of UAV, n_0 is power spectrum density of noise and b_u is the bandwidth between UAV-UAV links. To ensure that all data from users transmitted to UAVs can be successfully forwarded to their destinations, the SNR at each UAV should exceed a threshold γ_0. Based on the above inference, the value of SNR only depends on the distance between UAVs. Hence, to simplify the calculation, we alternatively define a communication range R_c of UAVs, so that any two UAVs with distance no bigger than R_c could establish a high quality link.

2.3.2. UAV-GU channel

To simulate the signal blocked by obstacles, the probabilistic LoS model in [40] is adopted for UAV-GU channel. In this model, the probability of LoS link between two nodes is defined with consideration to different environments, e.g., urban, suburban, rural areas, fields. Following the principle that city areas that have more buildings block more severely the signal than rural areas, the effect of different signal blockage can be modeled by adjusting hyper-parameters that formulate the probability of LoS and NLoS. At time slot t, the probability of LoS communication between GU $k^{(\cdot)}$ ($(\cdot)$ can be src or dst) and UAV m is given by

$$P^{\text{LoS}}_{k^{(\cdot)},m}[t] = \frac{1}{1 + ae^{-b(\theta_{k^{(\cdot)},m}[t]-a)}}, \tag{8}$$

where a and b are environmental constants on behalf of urban, suburban and rural places. $\theta_{k^{(\cdot)},m}[t]$ denotes the elevation angle between GU and UAV. Hence, the average path loss is

$$PL_{k^{(\cdot)},m}[t] = P^{\text{LoS}}_{k^{(\cdot)},m}[t] \times PL^{\text{LoS}}_{k^{(\cdot)},m}[t] + (1 - P^{\text{LoS}}_{k^{(\cdot)},m}[t]) \times PL^{\text{NLoS}}_{k^{(\cdot)},m}[t]. \tag{9}$$

The average path loss for LoS and NLoS are given by

$$\begin{cases} PL^{\text{LoS}}_{k^{(\cdot)},m}[t] = 20\lg(\frac{4\pi f_1 d_{k^{(\cdot)},m}[t]}{c}) + \eta_{\text{LoS}} \\ PL^{\text{NLoS}}_{k^{(\cdot)},m}[t] = 20\lg(\frac{4\pi f_1 d_{k^{(\cdot)},m}[t]}{c}) + \eta_{\text{NLoS}}, \end{cases} \tag{10}$$

where f_1 is the carrier frequency, $d_{k^{(\cdot)},m}[t]$ is the euclidean distance between GU and UAV, c denotes the speed of light. η_{LoS} and η_{NLoS} are the average additional path-loss under LoS and NLoS, respectively [41].

In the uplink part, we do not consider the interference from GUs to UAVs and assume this can be compensated utilizing the existing techniques, since GUs covered by the same UAV are severed with TDMA and this kind of interference is usually weak. The SNR generated at UAV m_1 by GU k^{src} is calculated by

$$\gamma^{\text{G2U}}_{k^{\text{src}},m_1}[t] = \frac{p^{\text{G}} \cdot h_{k^{\text{src}},m_1}[t]}{n_0 \cdot b_u}, \tag{11}$$

where p^{G} is the transmit power of GU, $h_{k^{\text{src}},m_1}[t] = 10^{-PL_{k^{\text{src}},m}[t]/10}$ is the channel gain. We assume the bandwidth are equally allocated to each GU, denoted by b_u. Therefore, at time slot t, the instant transmission rate of the uplink from GU k^{src} to UAV m_1 is calculated by

$$r_{k^{\text{src}},m_1}[t] = \begin{cases} b_u \cdot \log_2(1 + \gamma^{\text{G2U}}_{k^{\text{src}},m_1}[t]), & \gamma^{\text{G2U}}_{k^{\text{src}},m_1}[t] \geq \gamma_0, \\ 0, & \text{else}, \end{cases} \tag{12}$$

In the downlink part, the destination GUs receive data from UAVs. Since all UAVs operate at the same frequency when communicating with GUs, the interference at destination GUs from other UAVs should be considered. The signal to interference plus noise ratio (SINR) generated at GU k^{dst} by UAVs is calculated by

$$\gamma^{\text{U2G}}_{m_2,k^{\text{dst}}}[t] = \frac{p^{\text{U}} \cdot h_{m_2,k^{\text{dst}}}[t]}{I_{m_2,k^{\text{dst}}}[t] + n_0 \cdot b_u}, \tag{13}$$

where p^{U} is the transmit power of UAV. $I_{m_2,k^{\text{dst}}}[t]$ is the interference at destination GU caused by other UAVs that transmitting to other destination GUs, which is defined by

$$I_{m_2,k^{\text{dst}}}[t] = \sum_{j \in \mathcal{M}, j \neq m_2} \sum_{n \in \mathcal{M}} \beta^{dst}_{j,n} \cdot p^{\text{U}} \cdot h_{j,k^{\text{dst}}}[t]. \tag{14}$$

Therefore, at time slot t, the instant transmission rate in the uplink from UAV m_2 to GU k^{dst} is calculated by

$$r_{m_2,k^{\text{dst}}}[t] = \begin{cases} b_u \cdot \log_2(1 + \gamma^{\text{U2G}}_{m_2,k^{\text{dst}}}[t]), & \gamma^{\text{U2G}}_{m_2,k^{\text{dst}}}[t] \geq \gamma_1, \\ 0, & \text{else}, \end{cases} \tag{15}$$

where γ_1 is the SINR threshold of GUs.

Owing to the high quality links established between UAVs, in this paper, we assume that all data transmitted to UAVs can be successfully forwarded to their destinations. The bottleneck of transmission rate between the kth pair only depends on uplink or downlink transmission. At time slot t, the total number of GUs served by UAV m is

$$K_m[t] = \sum_{k \in \mathcal{K}} \alpha^{\text{src}}_{m,k}[t] + \beta^{dst}_{m,k}[t]. \tag{16}$$

Hence, the instant transmission rate between the kth pair is given by

$$r_k[t] = \tau^{\text{hover}}_m[t] \cdot \min(q_1 \cdot r_{k^{\text{src}},m_1}[t], q_2 \cdot r_{m_2,k^{\text{dst}}}[t]), \tag{17}$$

where $q_1 = \frac{\sum_{m_1 \in \mathcal{M}} \alpha^{\text{src}}_{m_1,k}[t]}{K_{m_1}[t]}$ and $q_2 = \frac{\sum_{m_2 \in \mathcal{M}} \beta^{dst}_{m_2,k}[t]}{K_{m_2}[t]}$.

3. Problem formulation

At time slot t, the accumulative throughput of the kth pair is calculated by

$$TP_k[t] = \sum_{t'=1}^{t} r_k[t']. \tag{18}$$

Then, the sum accumulative throughput of all pairs is calculated by

$$SUM_TP[t] = \sum_{k\in\mathcal{K}} TP_k[t]. \tag{19}$$

For fair communications, we define the measurement of fair throughput according to Jain's index [29] as follows

$$f[t] = \frac{\left(\sum_{k\in\mathcal{K}} TP_k[t]\right)^2}{K\left(\sum_{k\in\mathcal{K}} (TP_k[t])^2\right)}, \tag{20}$$

where $f[t] \in [\frac{1}{K}, 1]$.

For ease of presentation, the access policies of all GUs at time slot t are denoted by $\mathbf{A} = \{\alpha^{\text{src}}_{m,k}[t], \beta^{dst}_{m,k}[t]\}_{m\in\mathcal{M},k\in\mathcal{K},t\in\mathcal{T}}$, the locations of all UAVs are denoted by $\mathbf{W} = \{w^{\text{u}}_m[t]\}_{m\in\mathcal{M},t\in\mathcal{T}}$. Based on the above formula, our objective function is defined as

$$\begin{aligned} &S \max_{\mathbf{A},\mathbf{W}} \sum_{t\in\mathcal{T}} \frac{f[t]\cdot SUM_TP[t]}{\sum_{m\in\mathcal{M}} e_m[t]} \\ &s.t. \quad (1)\text{–}(4), \\ &C1: d^{\text{safe}} \le d_{m,n}[t], m \ne n, \forall m, n \in \mathcal{M}, t \in \mathcal{T}, \\ &C2: 0 < \lambda_2. \end{aligned} \tag{21}$$

where the numerator denotes the fair throughput of all pairs and the denominator denotes the energy consumption of all UAVs. Obviously, an increase in fairness or throughput and a decrease in energy consumption would both benefit the objective function, which exactly meets the system requirements. Eq. (1) constraints the movement of UAVs, while equations (2)–(4) constraint the access control of GUs. $C1$ imposes a hard constraint on the distance of two UAVs to avoid collision. $C2$ is the popular measurement of a graph named algebraic connectivity [42] denoted by λ_2, which is defined by the second smallest eigenvalue of the Laplacian matrix of a graph. When $\lambda_2 > 0$, the graph is connected, vice versa. Hence, $C2$ is to ensure that all UAVs form a connected topology when serving user pairs.

Due to the time-varying, discrete and continuous variables, the objective function is non-concave and intractable. DRL-based methods utilize deep neural networks (DNN) to extract policies that suitable for fair communications with designed reward function and can approximate the optimal distribution of UAV movements. Besides, MADRL encourages cooperation as well as interaction between agents and performs well in a lot of multi-agent system. As a consequence, in the next section, we propose a MADRL-based solution for multiple UAV trajectory design.

4. MADRL-based trajectory optimization for fair communications

4.1. IPPO preliminaries

DRL methods are mainly divided into two branches, the value-based and the policy-based methods. Value-based methods are used to solve discrete actions problems while the policy-based can solve continuous actions problems which is more suitable for our system. Traditionally, policy-based methods use an estimate function to update DNN parameters, which usually has the form as follows

$$\hat{g} = \hat{E}_t\left[\nabla \log(\pi_\theta(a_t|s_t))\hat{A}_t\right], \tag{22}$$

where $\pi_\theta(a_t|s_t)$ denotes the policy parameters, $\hat{A}_t$ is the estimator of the advantage function (difference between the normalized accumulative reward and the estimate of state value). However, in the process of updating network parameters, if we do not limit the difference between old and new policy, the policy may not be able to converge, which leads to lower efficiency in importance sampling.

TRPO (Trust Region Policy Optimization) [43], which aims to reduce the difference when updating policies, uses a surrogate objective as follows

$$\begin{aligned} &L^{CPI}(\theta) = \hat{E}_t\left[\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}\hat{A}_t\right] = \hat{E}_t\left[r_t(\theta)\hat{A}_t\right] \\ &s.t. \hat{E}_t\left[KL(\pi_\theta(\cdot|s_t), \pi_{\theta_{\text{old}}}(\cdot|s_t))\right] \le \xi, \end{aligned} \tag{23}$$

where $r_t(\theta)$ and $KL(\cdot)$ are the difference and divergence between new and old polices, respectively. ξ denotes the threshold of divergence. Based on the constrained updating mechanism, TRPO ensures monotonic convergence in the updating process.

PPO [44] is a policy-based DRL method integrating with actor–critic frame and also is the state-of-art algorithm. Different from TRPO that constrains the divergence, PPO utilizes a clipped surrogate objective function

$$L^{CLIP}(\theta) = \hat{E}_t\left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t)\right], \tag{24}$$

where ε is the clipping factor. $clip(\cdot)$ restricts the value of $r_t(\theta)$ between $1-\varepsilon$ and $1+\varepsilon$. In this way, the difference between new and old policies gets further reduced. The complete loss function of PPO is given by

$$L_t^{CLIP+VF+S}(\theta) = \hat{E}_t\left[-L_t^{CLIP}(\theta) + c_1 L_t^{VF}(\theta) - c_2 S[\pi_\theta](s_t)\right], \tag{25}$$

where c_1 and c_2 denote the loss factors, $L_t^{VF}(\theta)$ is the mean-square error (MSE) between the normalized accumulative reward and the estimate of state value, $S[\pi_\theta](s_t)$ is the entropy of policy distribution. With a batch-updating mechanism, PPO managed to get the optimal policy to choose action by minimizing the loss function above.

When we use PPO to train a multi-UAV model, it outputs joint actions of all UAVs directly, which is actually in a centralized way that needs global information. In order to leverage more the partial interaction between agents and the observable environment, MADRL methods have been effectively applied to complex and dynamic scenarios. Both IPPO [45] and MAPPO [46] is the extension of PPO into multi-agent system. MAPPO is implemented under the famous centralized training and decentralized execution (CTDE) structure as shown in Fig. 3(a). It updates critic network with global state and actions from other agents to reduce the random factors brought by the decisions of other agents and stabilize training. Even though the CTDE mode uses global state to update agents' critic networks and fasten the convergence, but it also increase the similarity of policies of agents which might cause bad performance. However, the author in [46] realized this problem and added an agent-specific information in the observation space of each agent to break the similarity, but the results are still not ideal. To break the similarity and encourage more exploration, IPPO directly treats the other agents as part of the environment and enables each agent updating its critic network according to its own observations without information from other agents, as shown in Fig. 3(b). As said in [45], *"Despite its various theoretical shortcomings, IPPO, a form of independent learning in which each agent simply estimates its local value function, can perform just as well as or better than state-of-the-art joint learning approaches"*. As a consequence, our solution is based on IPPO and will be introduced in the next subsection.

4.2. IPPO based multiple UAVs trajectory optimization algorithm

We model the flight of UAVs as partially observable MDP, since we assume each agent makes decisions according to their own observations in a distributed way. At time slot t, agent m has observation $\mathbf{o}_t^m$ and the global state is constructed as $\mathbf{s}_t = [\mathbf{o}_t^1, \dots, \mathbf{o}_t^m, \dots, \mathbf{o}_t^M]$. Take $\mathbf{o}_t^m$ as the input of actor network, we have $\mathbf{a}_t^m$. All actions from agents make up the joint actions $\mathbf{a}_t$ of the system, where $\mathbf{a}_t = [\mathbf{a}_t^1, \dots, \mathbf{a}_t^m, \dots, \mathbf{a}_t^M]$. Then,

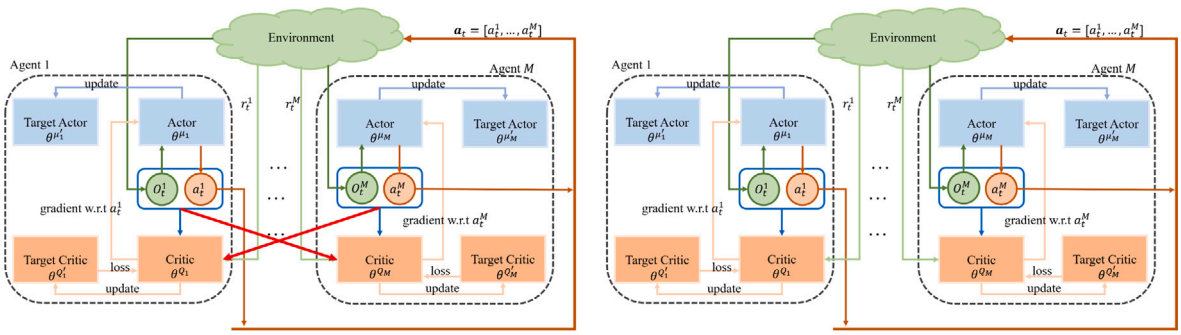


Fig. 3. The training framework of MAPPO and IPPO.

agents execute actions and get instant rewards $\mathbf{r}_t$ from the environment, while the state transits to $\mathbf{s}_{t+1}$ according to the transition probability matrix.

1) Observation Space: Each agent utilizes its observation to acquire its action. Hence, the observation space of each agent should contain effective information about the environment to guide the agent along the optimal trajectories. Take the mth agent as an example, the observation space contains the following items

$$\mathbf{o}_t^m \triangleq \left\{ \{w_m^{\mathrm{u}}[t]\}_{m\in\mathcal{M}}, \{w_k^{\mathrm{src}}, w_k^{\mathrm{src}}\}_{k\in\mathcal{K}}, \{r_k[t]\}_{k\in\mathcal{K}}, \{TP_k[t]\}_{k\in\mathcal{K}}, \{\alpha_{m,k}^{\mathrm{src}}[t]\}_{k\in\mathcal{K}}, \{\beta_{m,k}^{d,st}[t]\}_{k\in\mathcal{K}} \right\}. \tag{26}$$

It contains all UAVs' and GUs' positions, the accumulative throughput of each pair, the satisfaction of each pair as well as the access policies of each GU, which has the dimension of $[2M+8K]$. Remarkably, though the design of observation that contains all information seems to be centralized, agents actually use partial observations within their sense and communication range. Since the input dimension of neural networks cannot be dynamically changed during training, this definition aims to prevent execution error on in simulating and function like a placeholder.

2) Action Space: The action space of agent m is composed of moving direction $\theta_t^m \in [-\pi, \pi]$ and distance $l_t^m \in [0, l^{\max}]$. When $l_t^m = 0$, it will hovering for the whole slot. The joint action space of all agents is denoted by $\mathbf{a}_t = [\theta_t^1, l_t^1, \ldots, \theta_t^M, l_t^M]$, which has the dimension of $[2M]$.

3) Reward Function: This is the most important part of the model in RL, while proper reward design effectively gives correct feedback to agents and improper design might degrade the performance of algorithms. Our reward function is mainly composed of the cooperative part shared by all agents and the individual part.

At time slot t, the cooperative part of reward shared by all agents is calculated by

$$r_t^{\mathrm{share}} = f[t] \cdot \sum_{k=1}^{M} r_k[t], \tag{27}$$

where both long-term fairness throughput and short-term instant transmission rate are considered.

The individual part is composed of penalties and energy consumption. If an action chosen by the agent will exceed the border of the area, break safe distance or break the connectivity between UAVs, the agent will receive penalties and cancel this movement. The penalties of agent m at time slot t is given by

$$p_t^m = \alpha_m^{\mathrm{b}}(t) \cdot r^{\mathrm{b}} + \alpha_m^{\mathrm{s}}(t) \cdot r^{\mathrm{s}} + \alpha_m^{\mathrm{c}}(t) \cdot r^{\mathrm{c}}, \tag{28}$$

where r^{b}, r^{s} and r^{c} denote the border, safety and connectivity violations respectively. $\alpha_m^{\mathrm{b}}(t)$, $\alpha_m^{\mathrm{s}}(t)$ and $\alpha_m^{\mathrm{c}}(t)$ are 0–1 indicators. At time slot t, the complete instant reward agent m receives is calculated by

$$r_t^m = \kappa^{\mathrm{f}} \cdot \frac{r_t^{\mathrm{share}}}{e_t^m} - p_t^m, \tag{29}$$

where κ^{f} is the coefficient of fairness throughput and e_t^m is the energy consumption to stimulate energy efficiency behaviors. From the above reward function, we can tell that agents should cooperate with each other to increase the fair throughput to get higher rewards and each agent should reduce energy consumption as well as the violation of penalties to prevent the decrease of reward. Hence, the equivalence of the reward design to the objective function is proved.

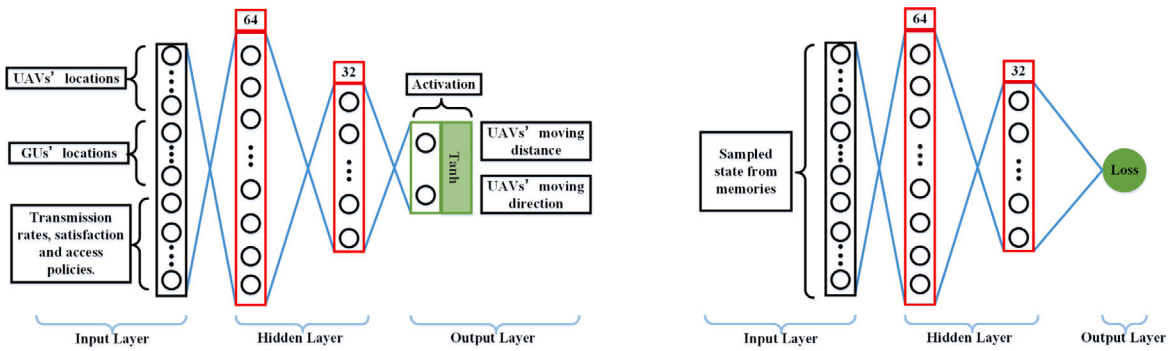
4) Actor–Critic Networks: Traditional CTDE uses global state and joint actions to update critic networks, which, to some extend, reduces the exploration and difference of agents due to the similarity of policies. In our network settings, each agent still possess their own actor–critic networks, and the difference lies in the absence of global information when updating critic parameters, only sampling from individual observation memories. In a word, a more distributed training mechanism is utilized in this paper, which intends to explore more actions and break the similarity of agents.

The actor network of each agent is shown in Fig. 4(a). The inputs is the agent's current observation, through two fully connected hidden layer and the output is a one dimension vector. All the activation functions used here are $Tanh()$. The only difference between critic and actor networks is that the input of critic is the sampled state from memories and the output changed to a loss value as shown in Fig. 4(b). In order to stabilize the training, we utilize an input-normalization mechanism that transfers the value of the raw input from each agent into a stable range.

5) Algorithms: The pseudo code of the training phase is shown in **Algorithm 1**. In the training phase, agent observes state from environment and makes decisions through actor network. Then, agent executes action, receives instant reward and inserts the information $(\mathbf{o}_t^m, \mathbf{a}_t^m, \mathbf{r}_t^m)$ into memory. After that, agent updates network parameters for several times with a certain period by calculating loss in (18). The only difference of MAPPO and IPPO lies in the update phase. MAPPO utilizes all information of all agents to update the network parameters while IPPO only allows each agent to utilize its own buffered information. The above process repeats until reaching maximum episodes. In the testing phase, agents directly use the well-trained model to output actions given observations without updating network parameters. The details of testing algorithm is the same as **Algorithm 1** after deleting Line 6 and Line 8, which has not been displayed in this paper for simplicity.

4.3. Algorithm analysis

The whole algorithm consists of training and testing phase. In both phases, DNN is the main part that contributes to the time complexity.



(a) Actor network (b) Critic network

Fig. 4. MADRL training networks design.

Algorithm 1 IPPO-based Multiple UAV Trajectories Optimization for Fair Communications—Training Phase

Initialization: Agents' memories; actor–critic networks parameters; update_batch K_B; max_episodes; episode = 0;

while episode < max_episodes **do**
  Reset environment and obtain an initial state $\mathbf{s}_0$.
  for $t < T$ **do**
    For each agent $m \in \mathcal{M}$, select $\mathbf{a}_t^m$ through actor network with $\mathbf{o}_t^m$ as input.
    Execute joint actions $\mathbf{a}_t$, get instant reward $\mathbf{r}_t$ and transit to next state $\mathbf{s}_{t+1}$.
    For each agent $m \in \mathcal{M}$, store $(\mathbf{o}_t^m, \mathbf{a}_t^m, \mathbf{r}_t^m, \mathbf{o}_{t+1})$ into its memory.
  end for
  For each agent $m \in \mathcal{M}$, calculates loss according to surrogate objective (23) and updates actor-critic network for K_B times with sampled data from memory.
  episode++
end while

Table 1
Parameters configurations.

Notations	Descriptions	Values
$e^{\mathrm{h}}, e^{\mathrm{f}}$	Energy unit for hovering and flying	5, 0.5
(a, b)	Probabilistic model constants	(9.61, 0.16)
f_1	Carrier frequency of UAV-GU channel	2 GHz
c	Speed of light	$3 * 10^8$ m/s
$\eta_{\mathrm{LoS}}, \eta_{\mathrm{NLoS}}$	Extra average path loss of LoS and NLoS	1, 20 dB
$p^{\mathrm{G}}, p^{\mathrm{U}}$	Transmit power of GU and UAV	10, 20 dBm
n_0	Power spectrum density of noise	−110 dBm/Hz
b_u	Bandwidth of UAV-GU channel	2 MHz
γ_0, γ_1	SNR threshold of UAV and GU	5, 5 dB
d^{safe}	Safe distance between UAVs	5 m
κ^{f}	Coefficient of fair throughput	20
$r^{\mathrm{b}}, r^{\mathrm{s}}, r^{\mathrm{c}}$	Penalties for violation of constraints	1, 1, 1

The action distribution is determined by two fully connected layers of the Actor in our model. The time complexity for DNNs is computed as the number of multiplications: $O(\sum_{f=1}^{F} n_f \cdot n_{f-1})$, where n_f is the number of neurons in layer f.

5. Simulation results

In this section, we conduct experiments on our proposed solution. The whole algorithm and simulation process are both implemented on a SuperMicro server with 4 NVIDIA TITAN V GPU based on Python scripts.

5.1. Experiment settings

The parameters used in our system are shown in Table 1. The mission area is set as a 2 km × 2 km square, while the mission duration $T = 200$ s is divided into 1000 slots with equal length $\delta = 0.2$ s. The altitude H of UAVs is 100 m and the maximal speed $V_{\max}$ is 40 m/s. In emergency scenario, like searching and rescue missions, where data needs to be transmitted to specific users and the relative position of the users in that pair may be randomly distributed due to the feature of the missions. Hence, at the start of each episode, GUs locations and pairing are randomly reset to catch the feature of any possible distributions.

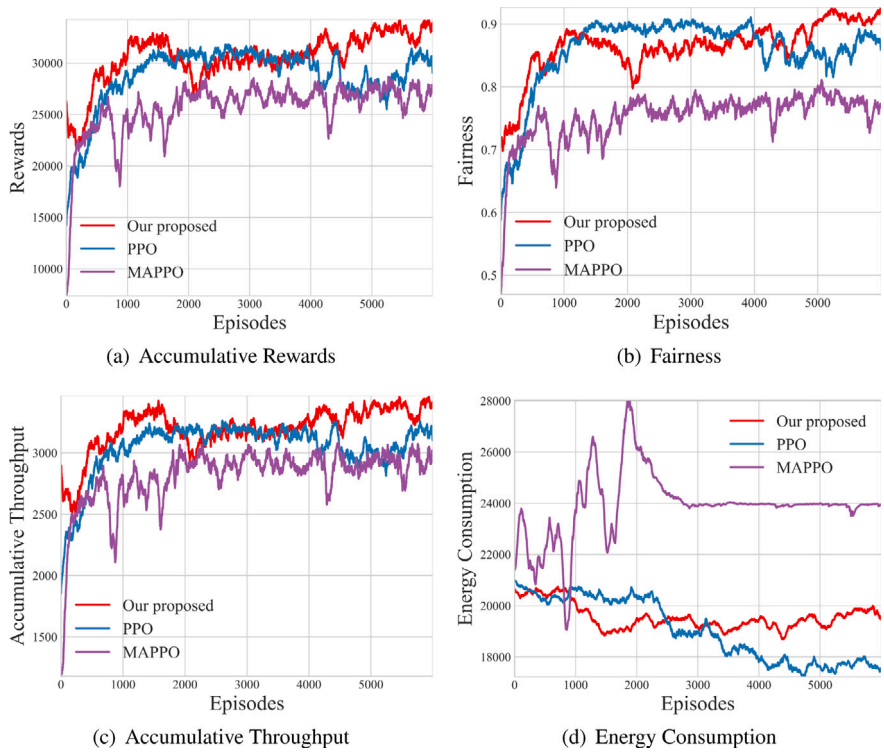
In the IPPO training phase, clipping factor $\varepsilon = 0.2$, loss function coefficient $c_1 = 0.5$ and $c_2 = 0.01$, discount factor $\gamma = 0.99$, learning rate for Adam optimizer is 0.003, and update_batch $K_B = 10$. Moreover, neurons in each hidden layer are set as 64 and 32 respectively.

5.2. Performance analysis

We compare our proposed solution with four baselines introduced below:

1. **Static K-means**: we use K-means to cluster the GUs into M clusters according to the distribution of GUs. Then, each UAV is deployed at the centroid of each cluster for the whole mission duration.
2. **Circle K-means**: we use K-means to cluster the GUs into M clusters according to the distribution of GUs. Then, each UAV is deployed to fly along a circular trace with the cluster centroid.
3. **MAPPO**: the source code in [46], which follows the CTDE mode uses global state as the input of critic networks.
4. **PPO**: the source code that ours extended from, where the joint actions of N UAVs are jointly generated.

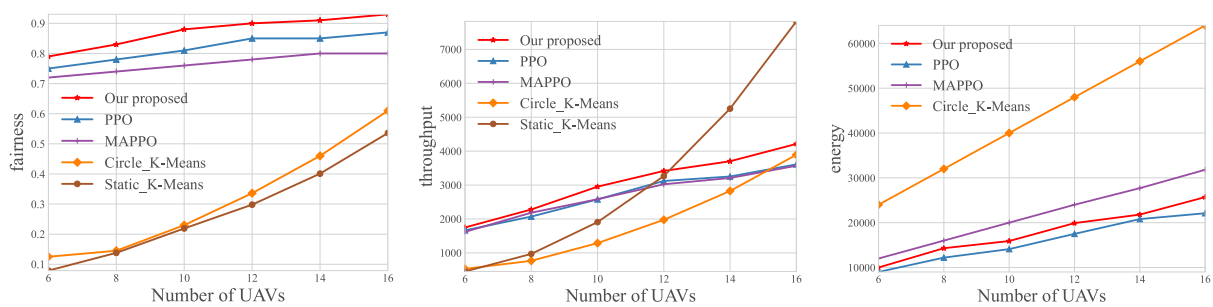
The convergence of the training phase is shown in Fig. 5. We train each model for 6000 episodes to see the trend of rewards, fairness, throughput and energy consumption. Note that at start of each episode, GUs locations and pairing are reset, which brings more dynamics by feeding adequate and dynamic user-pair distributions. Our algorithm outperforms the other two DRL method in accumulative rewards, fairness and accumulative throughput. MAPPO performs worst in all four metrics, which is due to the shared critic network parameters it utilizes that might make agents execute similar actions even though the situations are different. PPO has great performance in energy-saving over ours as shown in Fig. 5(d), which may result from the centralized decision. Ours consumes more energy to explore better actions and is still less than MAPPO. In other three metrics, PPO is close to ours. But at the later of the training phase, it starts to degrade and shows the instability, which might be caused by the centralized



(a) Accumulative Rewards (b) Fairness

(c) Accumulative Throughput (d) Energy Consumption

Fig. 5. Convergence of the training phase.



(a) Fairness (b) Throughput (c) Energy Consumption

Fig. 6. Impact of UAV number on the metrics.

way of choosing actions that requires more trainings to get its optimal results. However, in this simulation, we choose a lower cost with IPPO to get a similar performance. The gap of fairness between MAPPO and ours is the biggest while the throughput is the smallest, which indicates that updating with observations can ensure the diversity of policies to adapt to more dynamic environments. After the training converges and the model is well trained, agents use this model to sample actions according to their observations in the execution phase. The following simulation results are conducted on the well-trained model to evaluate its performance.

Then we conduct comparison experiments by considering the impact of different number of UAVs and GUs, respectively. The results of changing UAV number is shown in Fig. 6 with GU number fixed as 20 (communication pairs $K = 10$). In Fig. 6(a), the fairness metric gets higher with increasing UAV number for all algorithms. PPO has a similar trend as ours but smaller on the value. MAPPO's fairness is the smallest among the three DRL-based methods but higher than Circle and Static K-means. The curve of Static and Circle K-means shows great rising tendency, which is due to the clustering mechanism. As the number of clusters increases, the responsibility for each UAV decreases, then the fairness among pairs will also increases. But the fairness of Static is slightly lower than Circle K-means. In Fig. 6(b), the accumulative throughput increases as the number of UAVs increases. Circle K-means finally gets close to ours and exceeds the other two. This is because as the clusters increases, UAVs are closer to GUs, which promote the channel quality. However, Static K-means increases faster than Circle K-means and other DRL-based methods. This is because as the number of UAVs increases, there will be more hovering positions in the environment, and some of the pairs might have better communication links due to the static deployment of UAVs. Hence, the accumulative throughput would increase rapidly despite of lower fairness among different pairs. DRL-based methods grow slower due to the exploration for fairness communications. In Fig. 6(c), the energy consumption of Circle K-means is proportional to UAVs due to our experiment settings to fly along a circle all the time. The other three algorithms show the trend of increasing energy consumption as more UAVs deployed, where we can observe that PPO outperforms ours. This is because that PPO optimized the joint action of all UAVs to

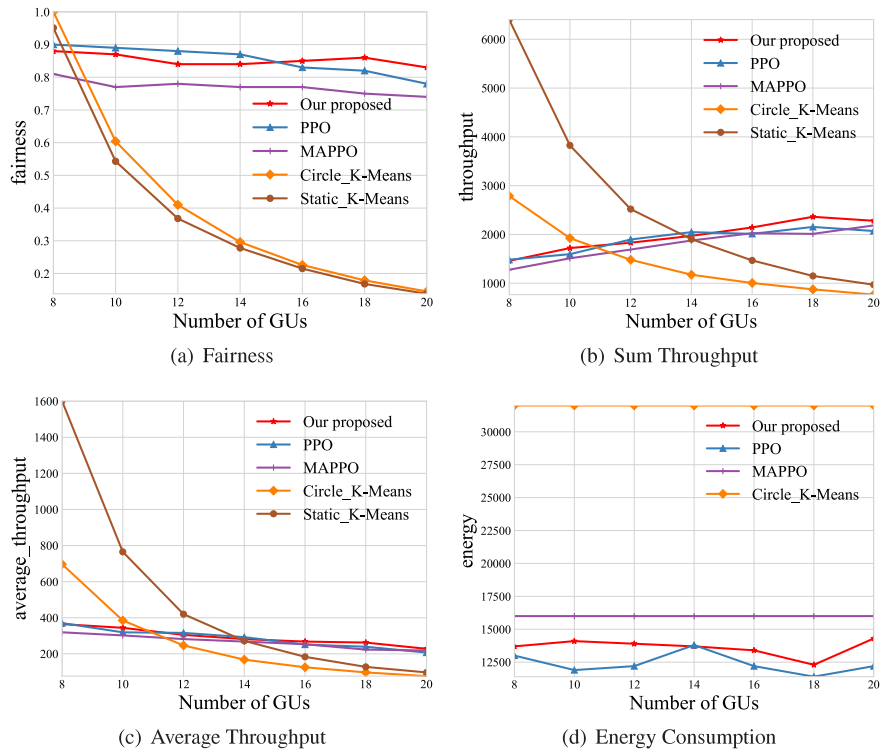


Fig. 7. Impact of GU number on the metrics.

maintain connectivity and provide fair service, which constrains on the movement of UAVs to some extend compared with IPPO. Moreover, connectivity and fairness are two coupled objectives. That is to say, ours pays more attention to fairness with the cost of connectivity level and guides UAVs to fly longer distance to serve.

The results of changing GU number (communication pairs) are shown in Fig. 7 with UAVs fixed as 8. Fairness decreases as more pairs need to be served in Fig. 7(a). However, ours shows the robustness when GUs exceed 12, as the others keep decreasing. In Fig. 7(b), the curves of sum throughput show the similar increasing trend for the three DRL-based methods, while both decreasing for the other two heuristic algorithms. Besides, when the number of GUs is smaller than 16, PPO is better than the proposed model. This is because that the mechanism of independent learning in ours encourages UAVs to explore more space and policies. Hence, when the number of GUs are relatively small and more exploration is not needed, ours performs worse than PPO. However, as the number of GUs exceeds the threshold, the benefit of exploration shows and ours outperforms PPO finally. In Fig. 7(c), the average accumulative throughput of each pair decreases as the number of pairs increases, which is to the common sense. Moreover, Static achieved more throughput than Circle K-means but paying the cost for lower fairness. In Fig. 7(d), the energy consumption shows a stable trend. The increase of GUs does not have an obvious effect on energy consumption.

Besides, Fig. 8 demonstrates the trajectories of UAVs of four algorithms with 6 UAVs and 20 GUs (i.e., 10 pairs). Our trajectories shown in Fig. 8(a) demonstrate the ability for fair communications and connectivity maintenance. Notably, UAVs have the intention to serve those far away from the center as they cooperatively reach different remote users. Circle K-means is the designed circular trajectories around the cluster centroids as shown in Fig. 8(b). In Fig. 8(c), though PPO also shows fair coverage, however, some UAVs have similar trajectories. Moreover, it reaches the boundary quickly and cannot response correctly when reaching the border (i.e., nothing but flying along the border), which is also the problem for MAPPO in Fig. 8(d). MAPPO performs the worst and generates almost the same straight traces for every UAVs. Obviously, the trajectories also indicate that our algorithm enables better fair coverage for multi-pair users than the other baselines.

6. Conclusion

We investigate a scenario that multiple UAV relays provide instant transmission service for multi-pair users for fair communications. In order to maximize the fair throughput of communication pairs and energy efficiency of UAVs, we propose a MADRL-based approach to learn the optimal policies that generate UAVs' trajectories. Meanwhile, we utilize algebraic connectivity as a constraint when choosing moving actions, which aims to maintain connectivity between UAVs to share more information for cooperation. We conduct experiments on our proposed solution together with three baselines. Based on the results and analysis, our algorithm outperforms these baselines in terms of fairness, throughput and energy efficiency.

CRediT authorship contribution statement

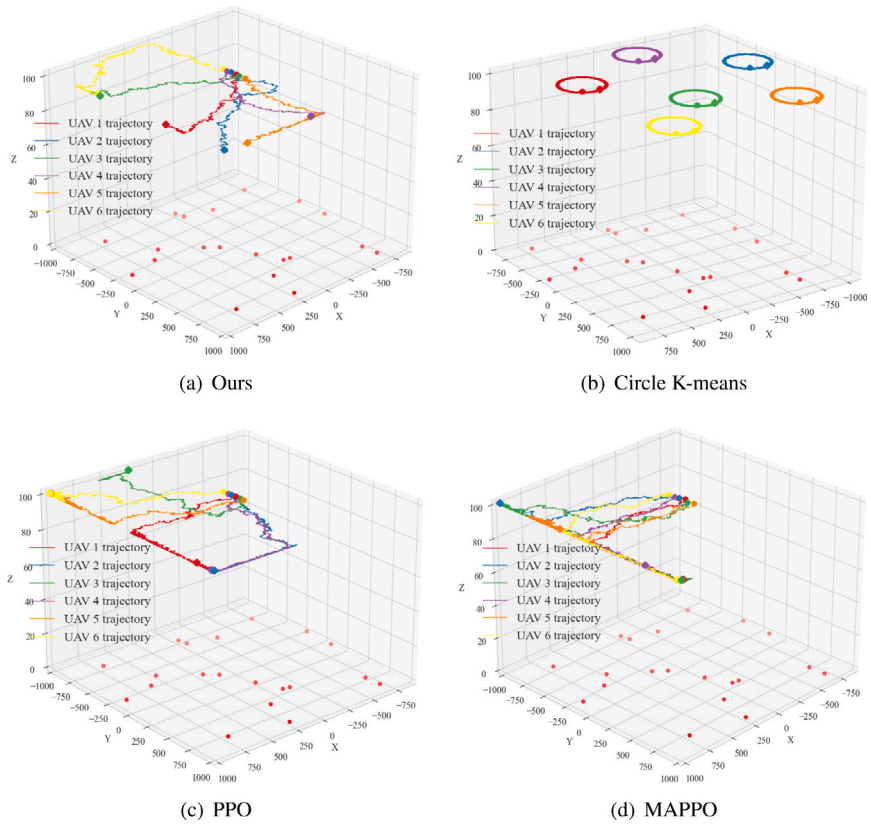
Xijian Luo: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Jun Xie:** Writing – review & editing, Methodology, Conceptualization. **Liqin Xiong:** Software, Methodology. **Zhen Wang:** Resources, Investigation. **Yaqun Liu:** Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.



(a) Ours (b) Circle K-means

(c) PPO (d) MAPPO

Fig. 8. Trajectories of UAVs with different algorithms.

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